Graphics Programming

Geometry Class

GB. Tony Liang

Geometry Class

- Store attribute data and total number of vertices
- Classes extending the base class add attribute objects to dictionary
- Two attributes (for now)
  - Vertex position
  - Default set of vertex colors

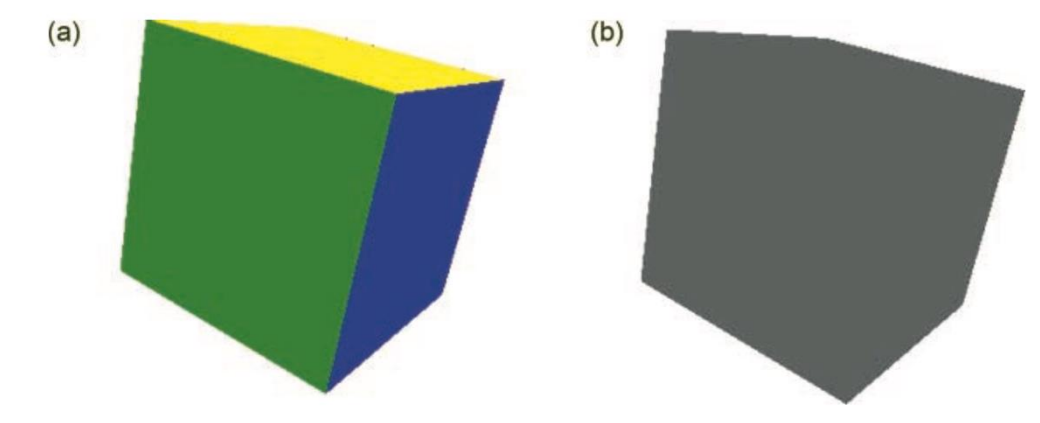
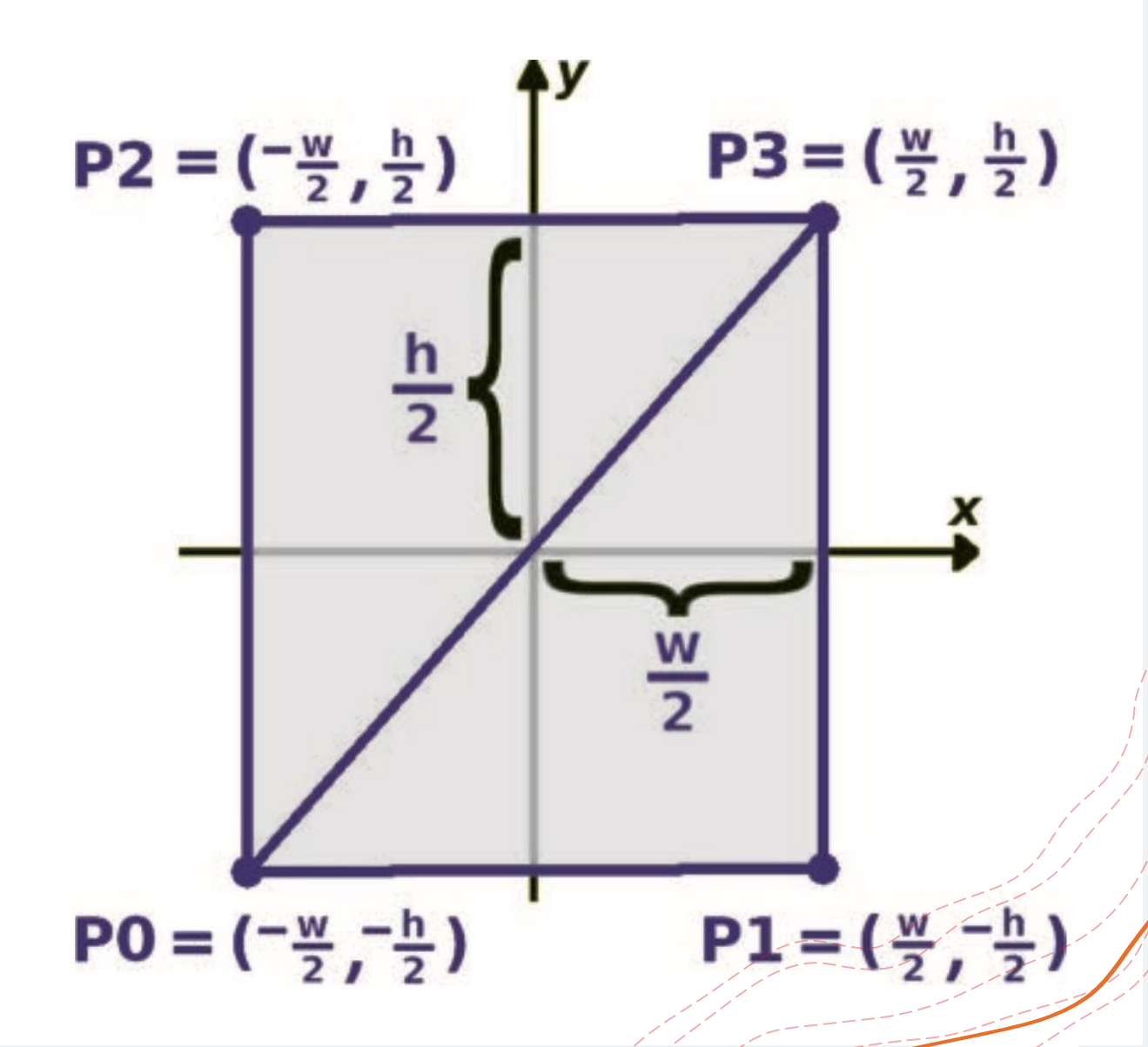


FIGURE 4.5 A cube rendered with (a) and without (b) vertex colors.

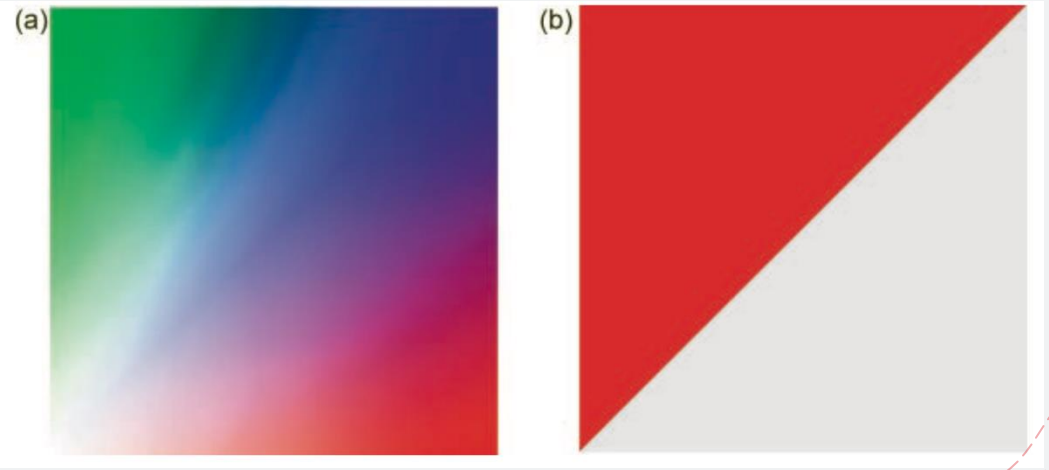
Rectangles – Vertex Positions

- Four vertices grouped into two triangles
- Constructor parameters:
  - Width and height of the rectangle
- Grouped into triangles
  - (P0, P1, P3)
  - (P0, P3, P2)



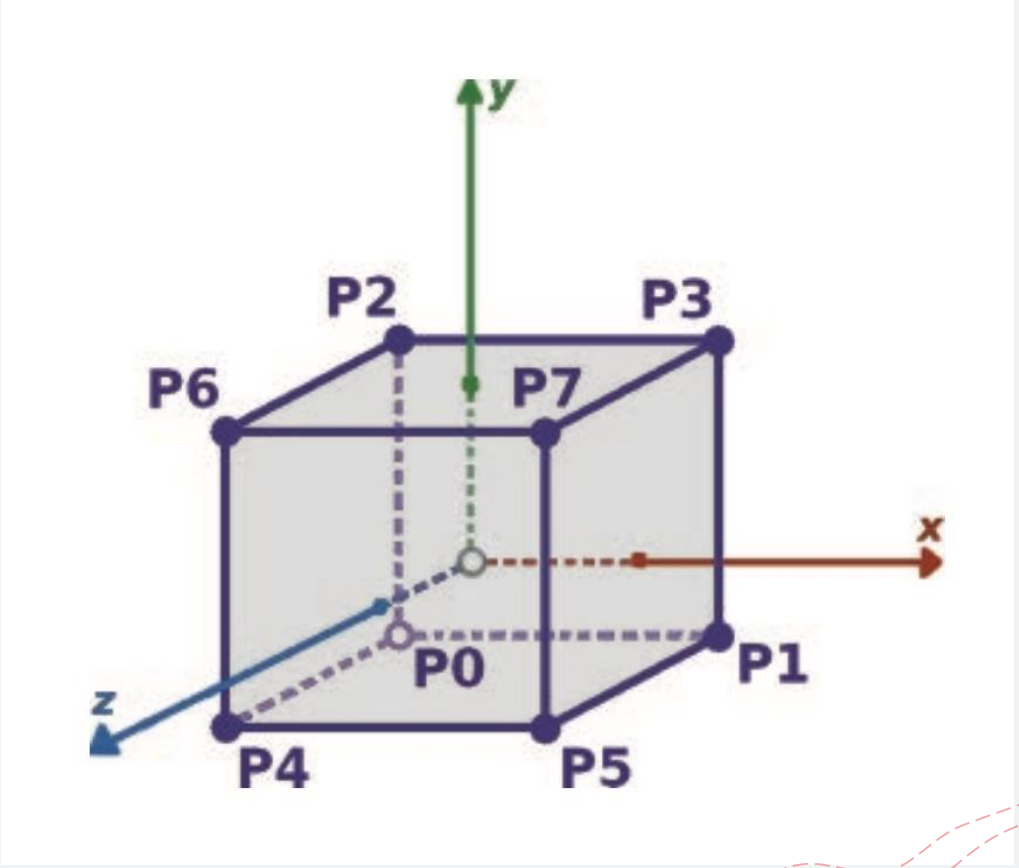
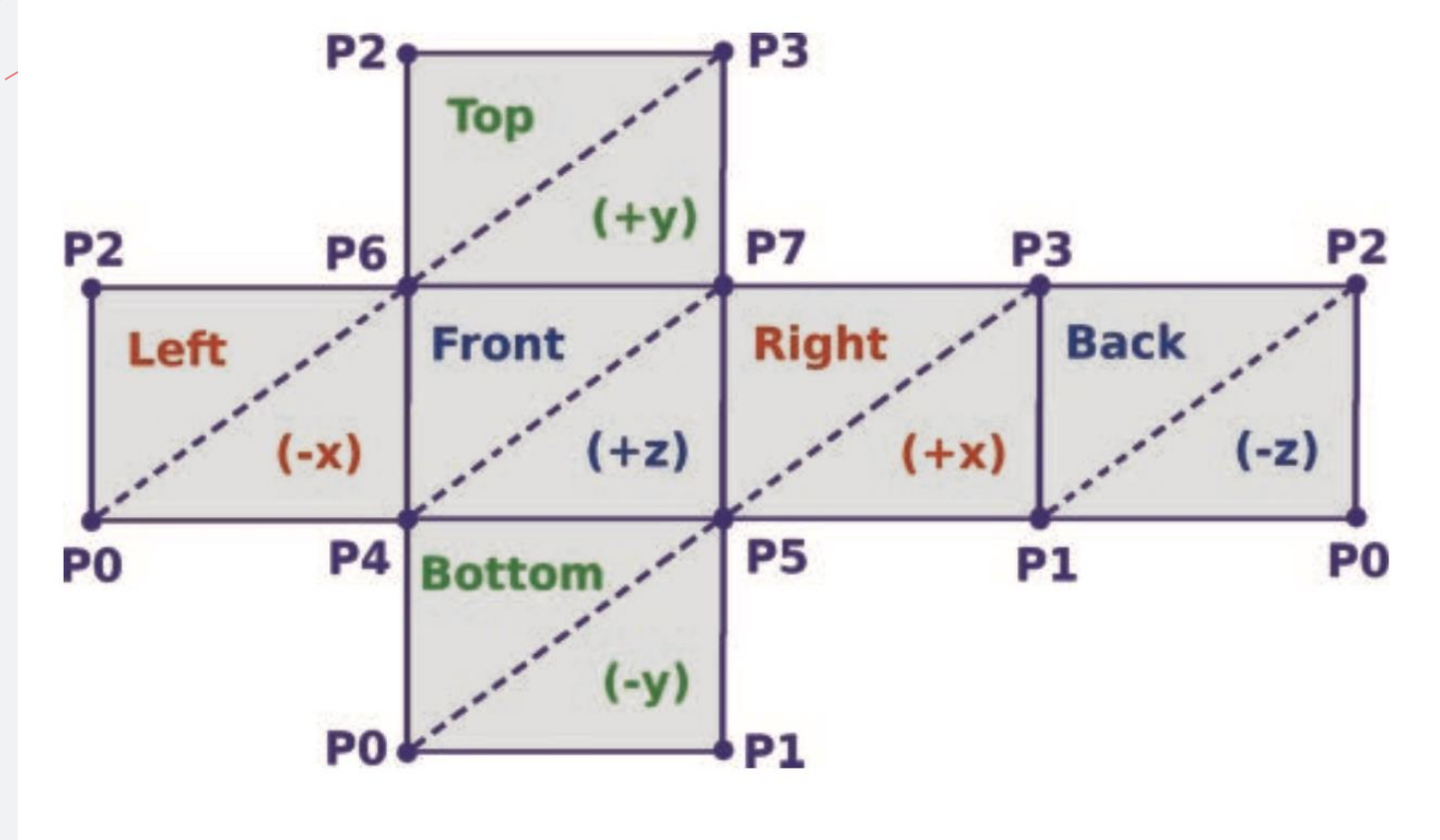
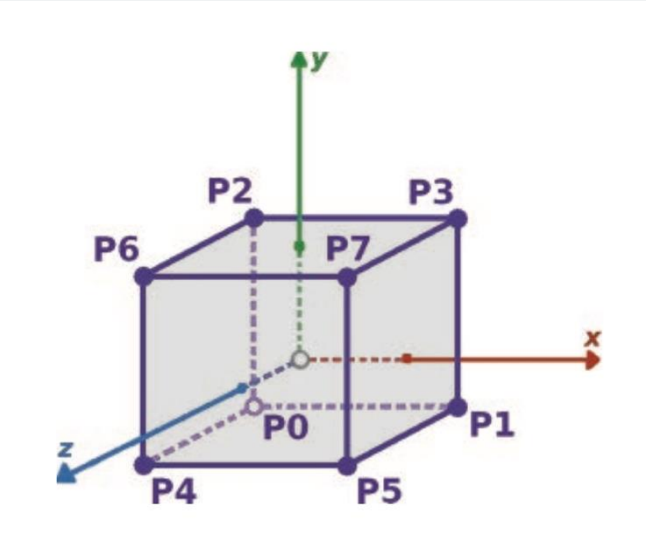
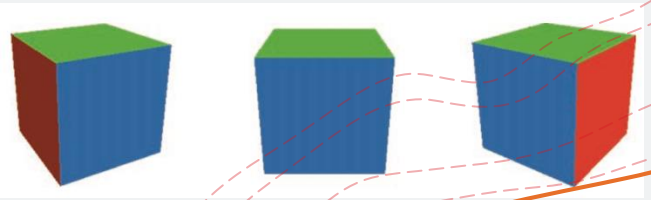
Rectangles – Vertex Colors

- Vertex color (C0, C1, C2, C3) listed in the same order as the position
- Create gradient effect
  - Recall: we have two groups of vertices
  - The order of vertex colors should follow the two groups
    - [C0, C1, C3, C0, C3, C2]
- Render solid color
  - [C0, C0, C0, C1, C1, C1]



A Box

+ 8 vertices
+ 6 rectangle sides (12 triangles)
+ Has width, height, and depth
+ Dished lines are blocked by the faces
+ Origin at the center of the box



Unfolding a Box